Does the sampler conserve what the protein conserves?

The zero_polymer chains against cross-subtype HA entropy

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Abstract

The sampler's conserved sites *are* significantly enriched among the sites influenza A hemagglutinin really conserves: the 53 K-sites of the T12 chain overlap the naturally conserved sites 3 times more often than chance, at $p \approx 10^{-7}$. But they are in the *wrong part* of the protein. The prediction written down before the data existed was that the sampler should lock the HA2 stalk; instead the K-sites are depleted in HA2 by a factor 3.4 and concentrated in the HA1 head, specifically around the receptor-binding site, where they are enriched 3.3-fold. The site-by-site rank correlations stay small throughout (0.06 to 0.10 at T12). The cross-subtype files are what made any of this measurable: on the old human-H3N2 file the same test has no power at all ($p = 0.27$), because 81% of that protein looks conserved.

1 What was run

Four natural entropy files, all built with `ha_site_entropy.py` from HA sequences fetched from NCBI Protein (16 subtypes, H1–H16; the counts are in Table 1):

file	subtypes	N_{eff}	NaN sites
`ha_group2_entropy_566.csv`	group 2 (6), $\theta = 0.8$	8.1	12
`ha_group2_w95_566.csv`	group 2 (6), $\theta = 0.95$	81.8	9
`ha_allA_entropy_566.csv`	all A (16), $\theta = 0.8$	15.7	16
`ha_allA_w95_566.csv`	all A (16), $\theta = 0.95$	234.3	16
`FASTA/H3N2/site_entropy_566.csv`	human H3N2 only	(unweighted)	0

against the sampled per-site entropy already in `nsites_study/structure/site_entropy_by_T.csv`: T5 and T12 (one chain each), T32 (two chains pooled) and T100 (four chains).

2 The data behind the natural files

Two things in this table matter for reading the result.

The identities confirm the mapping worked. H3 comes out at 0.959 to the reference, as it should, since the reference is an H3. H4 at 0.650 and H14 at 0.643 are the next closest, which is exactly the group 2 structure expected from the phylogeny, and every group 1 subtype sits near 0.40. Nothing is mis-mapped.

At $\theta = 0.8$ the weighting over-corrects. The N_{eff} column at $\theta = 0.8$ (not shown above; in the `_entropy_566.summary.tsv` files) is 1.0 for most subtypes, as expected, but 0.0 for H15, 0.4 for H16 and 0.7 for H13. Those are not bugs: sister subtypes are more than 80% identical to each other, so H15 (11 unique sequences) is absorbed into the H7 cluster (1 712 sequences)

Table 1: Per-subtype composition of the all-A files, from `ha_allA_w95_566.summary.tsv`. "Unique" counts distinct sequences after filtering; `-max-unique` caps each subtype at 3 000, which bound H1, H3 and H5 only. Identity is the median over that subtype to the `zero_polymer` reference, after alignment.

subtype	raw	unique	kept	N_{eff} at 0.95	identity to ref
H1	28 717	13 541	3 000	66.1	0.417
H3	26 377	11 617	3 000	42.7	0.959
H5	6 718	4 741	3 000	23.5	0.399
H9	3 678	2 707	2 707	22.1	0.392
H7	3 201	1 712	1 712	19.3	0.473
H6	1 821	1 408	1 408	16.3	0.403
H4	2 012	1 157	1 157	6.2	0.650
H10	1 170	622	622	5.8	0.482
H11	665	590	590	6.3	0.405
H2	629	483	483	5.1	0.389
H13	509	258	258	8.4	0.380
H12	223	201	201	2.1	0.403
H16	262	141	141	6.0	0.376
H8	161	151	151	1.1	0.403
H14	30	20	20	1.3	0.643
H15	22	11	11	2.0	0.475

and ends up with essentially zero total weight, and H13/H16 absorb each other. So at 0.8 the entropy is not "one vote per subtype" as the runbook suggests, it is one vote per *cluster of sister subtypes*, with small subtypes next to large ones silently dropped. **Prefer the $\theta = 0.95$ files**; this was not something I anticipated when writing the method, and it is the one correction to make to `ha_natural_methods.tex`.

3 Result 1: the natural files behave as predicted

The prediction was HA2 conserved, HA1 head variable. On the natural side it holds, and more cleanly in the cross-subtype files than in H3N2 (median per-site entropy, bits; signal peptide 0–15, HA1 16–344, HA2 345–565, the HA2 boundary located from the fusion peptide motif `GIFGAIAGF` at index 345):

file	signal	HA1	HA2
group 2, $\theta = 0.8$	1.993	1.478	0.812
group 2, $\theta = 0.95$	2.092	1.276	**0.454**
all A, $\theta = 0.8$	2.704	1.809	0.991
all A, $\theta = 0.95$	2.616	1.709	1.060
human H3N2	0.049	0.015	0.007

HA2 is about a third of HA1's entropy in the group 2 file at $\theta = 0.95$, and the signal peptide is the most variable region of all, which is correct: it is cleaved off and only its hydrophobicity is under selection. The pipeline is measuring what it was built to measure.

Look at the H3N2 row, though. Its numbers are 20 to 100 times smaller, because within one lineage almost nothing varies at all. That is the quantitative version of the argument for doing this work, and Section 5 is where it bites.

4 Result 2: the sampler conserves real sites, in the wrong place

The K-sites are the sites with sampled entropy below 0.05 at the cold labels. **Only T12 is usable.** The T5 chain accepted 179 mutations over 566 sites, 0.32 per site, so its 518–520 "conserved" sites are overwhelmingly sites the chain never touched, not sites it refused to change. T12 accepted 26 666 mutations over 1 001 kept checkpoints, 47 per site, so its 53 K-sites mean something. Everything below is T12.

The overlap is real. Against the group 2 $\theta = 0.95$ file, 23 of the 53 K-sites are among the 83 sites with natural entropy below 0.1 bits, where chance would give 7.9: $p = 9.9 \times 10^{-8}$. The same test on the other files gives $p = 2.9 \times 10^{-6}$ (group 2, 0.8), 6.2×10^{-6} (all A, 0.95) and 6.7×10^{-5} (all A, 0.8). The median natural entropy at the K-sites is 0.338 bits against 1.076 at the remaining sites. So the sites the sampler refuses to change are, as a set, genuinely unusually conserved in nature. This is the positive result, and it is not marginal.

But they are in the head, not the stalk.

region	sites	K-sites	expected if uniform	enrichment
signal (0–15)	16	1	1.5	0.67
HA1 (16–344)	329	**46**	30.8	**1.49**
HA2 (345–565)	221	**6**	20.7	**0.29**

This is the opposite of the prediction. HA2, the region the natural alignment says is the most constrained part of the molecule, is where the sampler is *least* likely to hold a site fixed. Of the 6 K-sites that are in HA2, only 1 is naturally conserved. Within HA2 alone the rank correlation between sampled and natural entropy is 0.074, i.e. nothing.

The one pleasing detail is that index 345, the very first residue of HA2 and so the first residue of the fusion peptide, *is* a K-site. The sampler found the single most famous constrained residue in the molecule and then ignored the rest of the stalk.

What the HA1 K-sites are. They are not scattered through the head. Sixteen of the 53 fall inside the classic receptor-binding neighbourhoods (H3 numbering 130–140, 150–160, 183–200, 219–229), which are only 52 of 566 sites, so chance would give 4.9: a 3.3-fold enrichment. In H3 numbering the K-sites include an almost unbroken run 190–200 (190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200) and a second cluster 239–263. The 190-helix is a core element of the receptor-binding site.

Three checks that this is not noise:

- **Spatial clustering.** The 53 K-sites form 36 contiguous runs; 53 sites placed at random form 48.1 ± 2.0 runs, so the clustering is far beyond chance ($P \leq 36$ is below 10^{-4} over 20 000 draws). The sampler is freezing structural neighbourhoods, not isolated positions.
- **Temperature consistency.** The T32 chains, a different temperature and two pooled chains, give 13 K-sites, and 12 of those 13 are also K-sites at T12. The cold-chain conserved set is reproducible and not an artifact of one trajectory.
- **Mutational opportunity.** At T100 no site at all has entropy below 0.05, so the proposal mechanism does reach every site; a frozen site at T12 is a rejection, not an absence of attempts.

The rank correlations are small everywhere.

natural file	T5	T12	T32	T100
group 2, $\theta = 0.8$	0.124	0.063	0.014	0.038
group 2, $\theta = 0.95$	0.120	0.077	0.035	0.050
all A, $\theta = 0.8$	0.122	0.083	0.029	0.025
all A, $\theta = 0.95$	0.145	0.104	0.025	0.020
human H3N2	0.131	0.086	−0.040	0.023

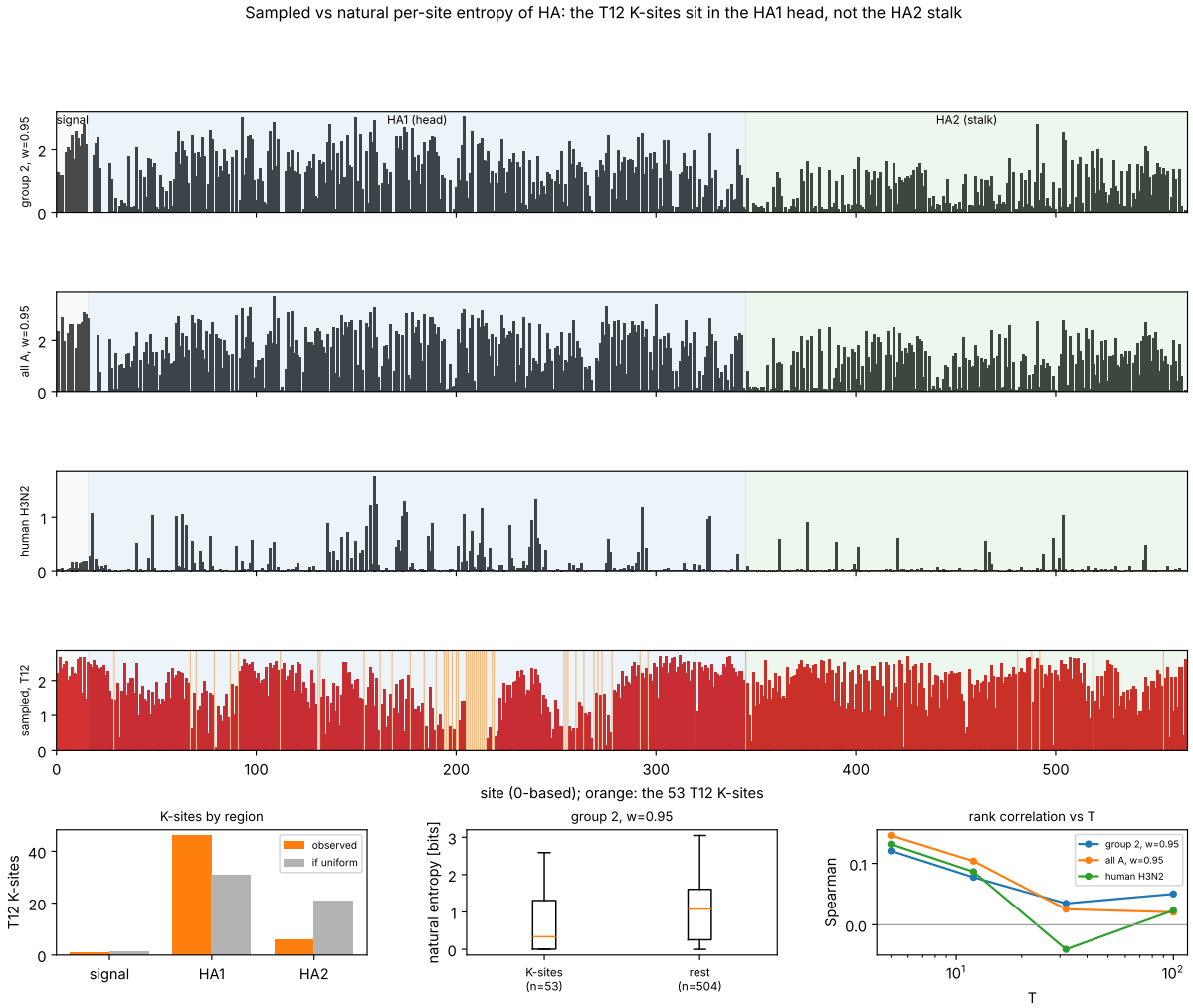
Do not read the T5 column: as above, that entropy is mostly an artifact of too few accepted mutations, and its apparently "best" correlation is the least meaningful number in the table. At T12 the honest figure is 0.06–0.10. The correlation falls with temperature, which is at least the right direction: the hotter chains look less like the natural protein. But a site-by-site correlation of 0.08 is weak, and the result has to rest on the K-site overlap, which is a statement about the extremes of the distribution, not about the distribution as a whole.

5 Result 3: this is why the cross-subtype files were needed

On the human-H3N2 file, 457 of 566 sites have entropy below 0.1 bits, i.e. 81% of the protein counts as "naturally conserved". The K-site overlap test then gives 45 observed against 42.8 expected, $p = 0.27$: no signal, because almost any set of 53 sites would score the same. The cross-subtype files cut "conserved" down to 72–131 sites out of 566, and the same 53 K-sites then come out at $p \approx 10^{-7}$.

So the headline positive finding in this note is one that could not have been obtained from the old file. The comparison was not merely noisier before, it was statistically empty.

6 Figure



Top three panels: natural per-site entropy for group 2 and all-A at $\theta = 0.95$ and for human H3N2, on the same site axis, with the signal peptide, HA1 and HA2 spans shaded. Fourth panel: the sampled entropy at T12, with the 53 K-sites as orange lines; the visible clusters of orange in the middle of HA1 and their near-absence beyond site 345 are the whole result. Bottom row: K-sites by region against the uniform expectation, natural entropy at K-sites against the rest, and the rank correlation against temperature.

7 How to read this, and what I would do next

The sampler is not reproducing nothing. It holds a set of sites that is significantly biased towards genuinely conserved positions, it does so reproducibly across temperatures, and it freezes contiguous structural neighbourhoods rather than scattered residues. That is more than a null result.

But it is conserving the *static folded core of the head*, above all the receptor-binding region, and not the *stalk*. There is a plausible physical reason, and it is a statement about the energy, not about the sampling: U is a distance between ESM3 attention maps. Attention maps report on the contacts that define a rigid, well-packed structure, which the β-sheet head core and the receptor-binding site are. The HA2 stalk's natural conservation is of a different kind: it is conserved because it must be *metastable*, able to undergo the large pH-triggered rearrangement that drives membrane fusion. Nothing in a distance between attention maps of the pre-fusion

state can see a constraint whose content is "this structure has to be able to become a different structure". If that reading is right, the mismatch is a property of U_{am} as a foldability proxy, and no improvement to the move set will fix it.

Concretely, in order of how much they would change the story:

1. **Check the energy, not the sampler.** Compute U for the reference with single mutations placed (a) at HA2 stalk sites that are conserved naturally but free in the sampler, and (b) at the HA1 K-sites. If the stalk mutations really do cost little, the mismatch is in U and this becomes a finding about the energy function. That is a cheap, decisive experiment and I would do it first.

2. **Run the comparison on the finished T12 chain.** The numbers here use the 1 001 checkpoints that existed when the entropy file was written. The T12 chain has since reached 1M moves; redoing the entropy on the full post-burn-in set may move the K-site count.

3. **Use the secondary tier.** The 0.05 nat threshold gives 53 sites. The paper's "lower than 1" tier would give a larger set and a less extreme test; if the HA1/HA2 asymmetry survives there too, it is not a threshold effect. A sensitivity run at 0.01, 0.05, 0.1 and 0.3 is still outstanding.

4. **Fix the $\theta = 0.8$ over-correction** noted in Section 2, or simply standardise on $\theta = 0.95$ and say why.

One caveat on all of it. Every number here comes from a single cold chain, T12. The temperature consistency check with T32 is reassuring but T32 is above the transition. Before any of this goes in the thesis it wants a second independent cold chain, which is what the T5 resume was for; at 0.32 accepted mutations per site T5 cannot yet serve that purpose.